Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name:

Team leader: Jayesh Rajani

Problem statement: Fast, local crop triage for farmers who cannot wait for an agronomist.

Your crop can’t talk. We listen anyway.

Brief about the idea

A second opinion for a sick crop—built from more than a picture.

RootCause combines a leaf photo, a spoken symptom note and field conditions to surface likely causes, explain the evidence, show what is missing and suggest safe next steps.

Triage, not a verdict.

The farmer keeps the final say—and the KVK expert stays in the loop.

The opportunity

CONTEXT

Photo + voice + weather + satellite + labelled references.

EVIDENCE

Each cause earns a 0–100 fit with reasons for and against.

SAFETY

Weak fits stay hidden; generic actives and an expert check guide treatment.

The sky, the soil canopy and the farmer’s own words all get a vote.

A local-language second opinion; a nearby shop only after the evidence and safety guidance.

What RootCause does

01

REPORT

Photo-first. Speak in Hindi or English. Confirm the crop only when needed.

02

INVESTIGATE

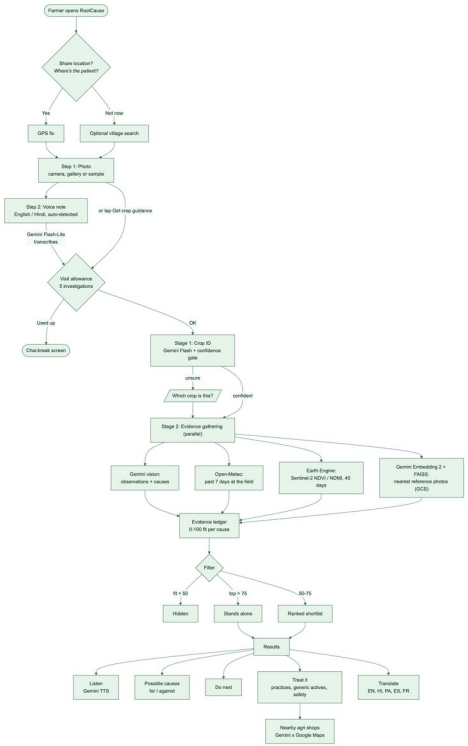
Cross-check the leaf with references, local weather and Sentinel-2.

03

ACT

See likely causes, urgency, safe next steps and nearby agri shops.

From a leaf photo to a careful next step



01 **REPORT**

Photo + voice; location optional.

02 **IDENTIFY**

Crop ID asks when confidence is low.

03 **CROSS-CHECK**

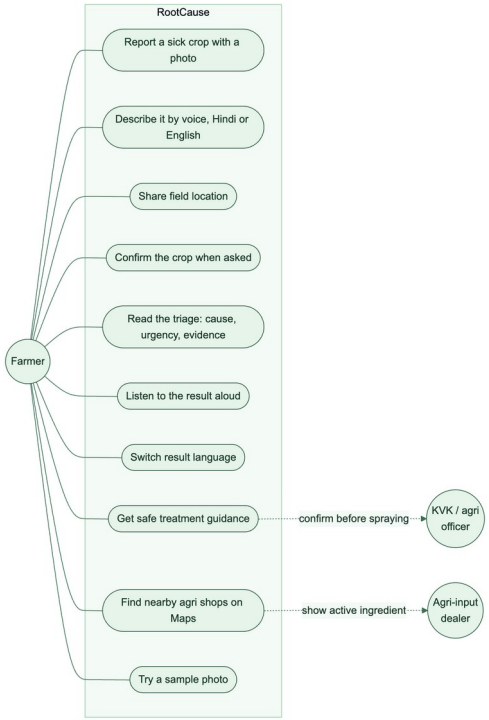
Weather, satellite, references and vision run together.

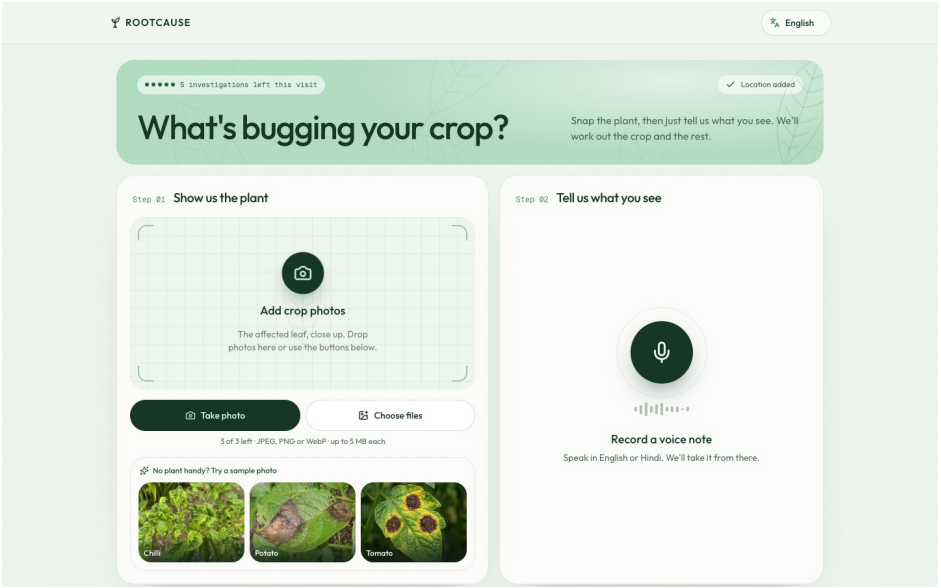
04 **RESPOND**

Ranked causes, next steps, treatment and shops.

No signal is treated as normal when it was not checked.

Designed around the farmer

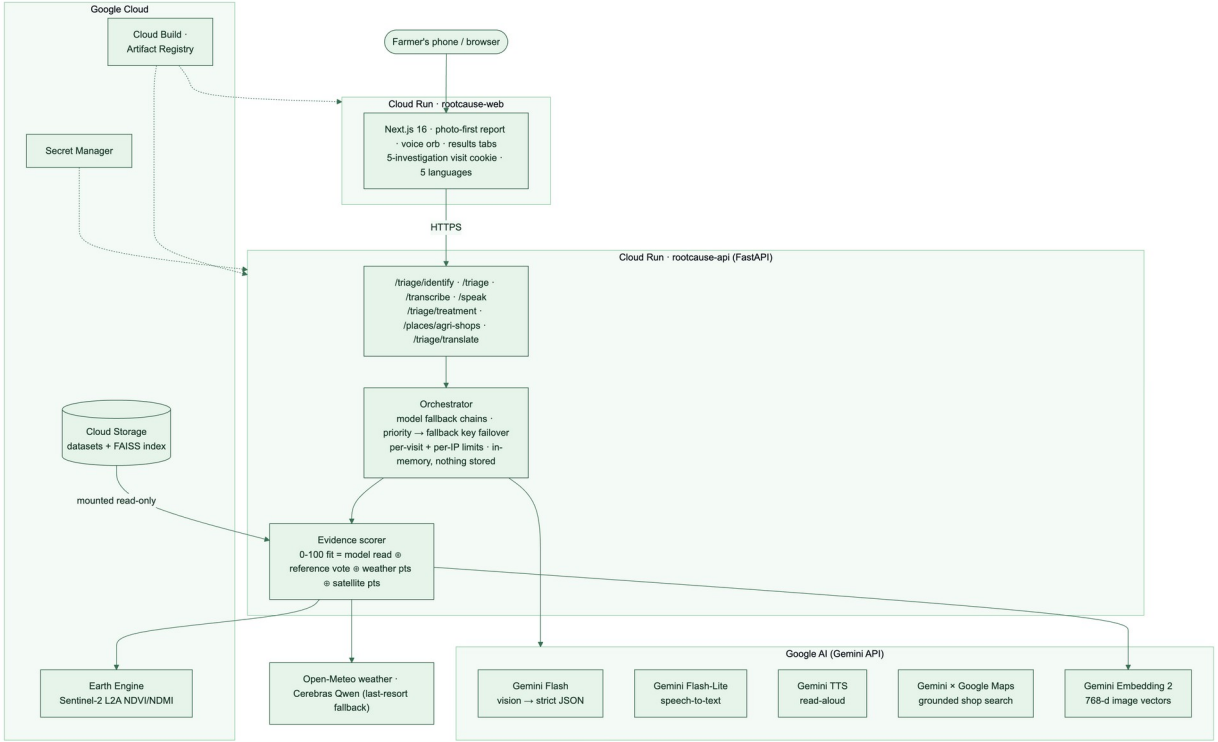




FROM FIRST PHOTO TO PRACTICAL NEXT STEP

Optional location. Spoken or typed. Clear next steps.

System architecture



Technology stack

GOOGLE AI

Gemini Flash · triage and vision
Gemini Embedding 2 · references
Flash-Lite · transcription
Gemini TTS · read-aloud
Gemini + Maps · real shops

GOOGLE CLOUD

Cloud Run · web + API
Cloud Build + Artifact Registry
Cloud Storage · FAISS library
Secret Manager · API keys
Earth Engine · Sentinel-2

FIELD + APP

Open-Meteo · local weather
Cerebras · last-resort model
Next.js 16 · React 19
FastAPI · Pydantic schemas
Five result languages

Prototype cost

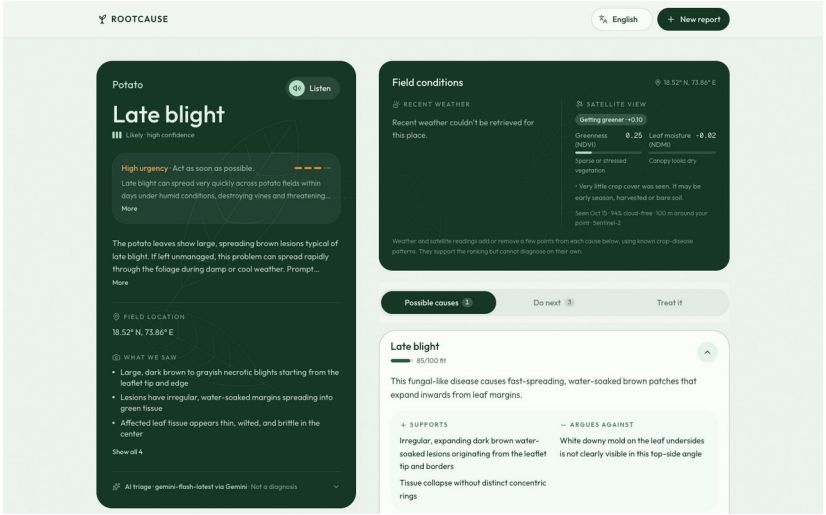
≈ $5

ESTIMATED PROTOTYPE CASH SPEND

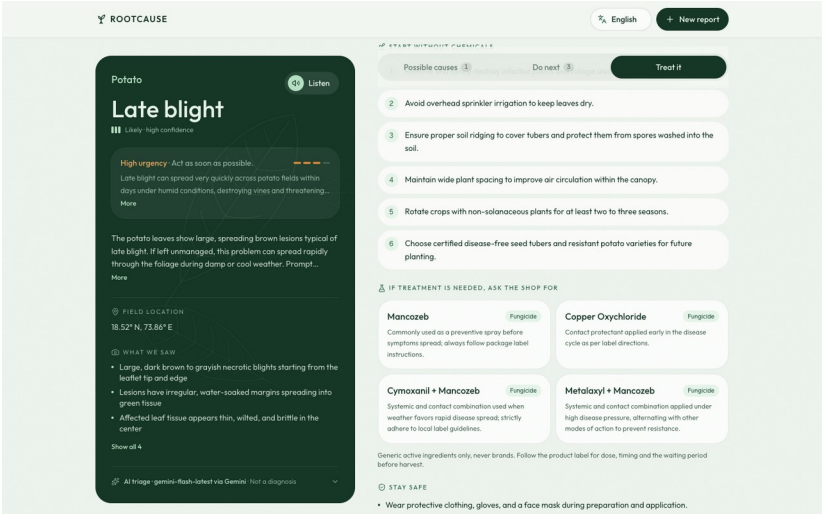
Gemini prepaid credit: $5
Cloud Run: ~$0 at demo traffic (scale-to-zero)
Cloud Storage: < $0.10/month
Earth Engine + Open-Meteo: $0 for this prototype use
Build, registry and secrets: within free allowances at demo volume

Planning estimate from the submission README; actual usage charges may vary.

A real product, not a mock-up



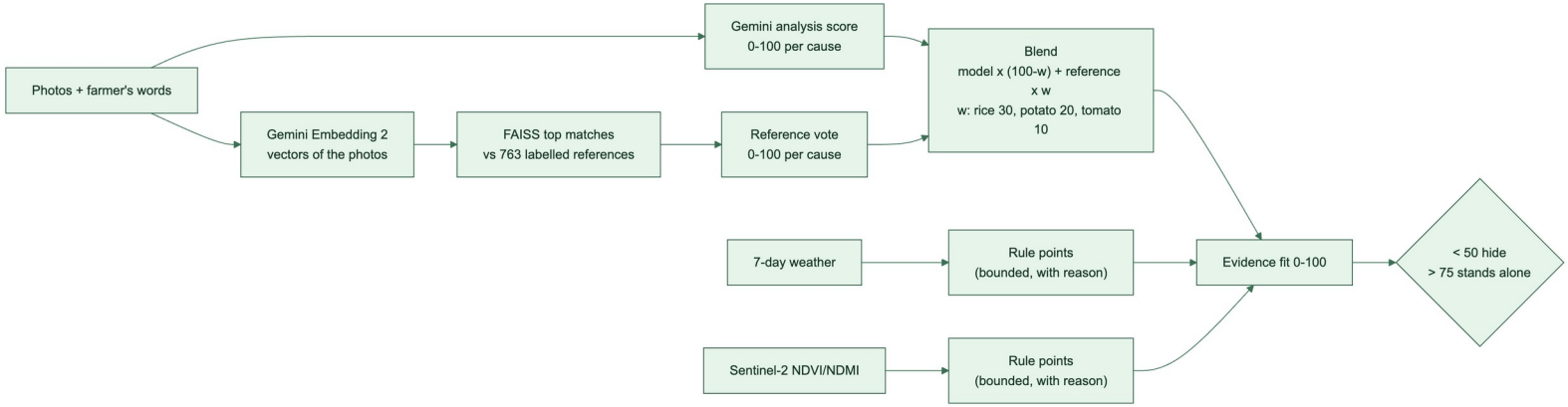
EVIDENCE BEFORE CERTAINTY



PRACTICAL, SAFER ACTIONS

Causes show supporting and contradicting evidence. Treatment opens on demand and keeps the expert check visible.

Prototype performance



763

labelled reference images

333

held-out evaluation images

25–30 s

full investigation (prototype)

Retrieval metrics in the README are DINOv2 baseline results; Gemini Embedding 2 production-index evaluation is not reported yet. Rice top-cause hit rate improved 0.50 → 0.90 on 10 held-out field photos.

What comes next

ASK BACK	A short voice question when two causes are close.
WARN EARLIER	Use the five-day forecast for disease-risk alerts.
CONNECT PEOPLE	Opt-in district signals and shareable KVK case cards.
GROW COVERAGE	More crops, regions and offline first-pass triage.

A diagnosis should lead to an expert when it needs one.

Provide links to your:

PUBLIC REPOSITORY

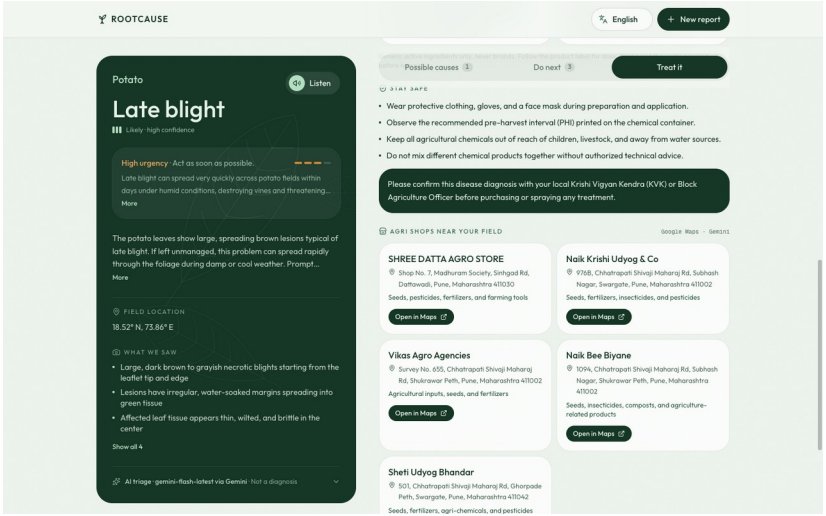
https://github.com/jayeshrajani/RootCause

LIVE PRODUCT

https://rootcause-web-lp7bmse7yq-el.a.run.app/

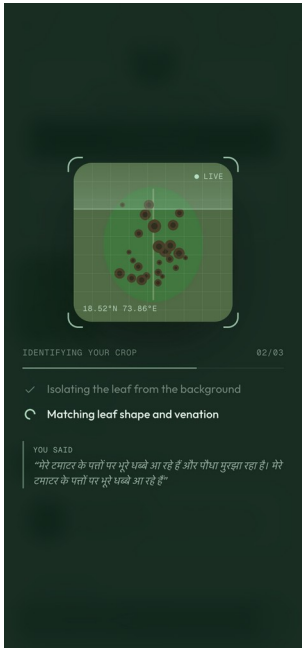
DEMO VIDEO

A few more things the farmer can do



REAL PLACES, OPEN IN MAPS

Photos and audio are processed in memory and never stored or logged.



STEPS SHOWN AS THEY RUN

Google Cloud H2S

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Thank you

